AP CALCULUS AB

Unit 1 • Lesson 1.12 notes

Lesson 1.12 • Confirming Continuity over an Interval

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Identify intervals on which common functions are continuous.
- Use the continuity of sums, products, quotients and compositions.

Vocabulary. continuous on an interval, one-sided continuity, domain.

1. Continuous on an interval

f is continuous on an open interval if it is continuous at every point of it; on $[a, b]$ we also require $\lim_{x \to a^+} f(x) = f(a)$ and $\lim_{x \to b^-} f(x) = f(b)$.

Functions continuous on their domains

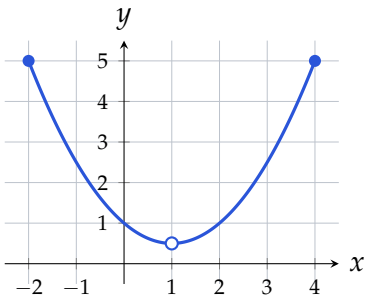
Polynomials (all reals); rational functions (except zeros of the denominator); $\sqrt{x}$ on $[0, \infty)$; e^x (all reals); $\ln x$ on $(0, \infty)$; $\sin x, \cos x$ (all reals); $\tan x$ except at $\frac{\pi}{2} + k\pi$. Sums, products, quotients (nonzero denominator) and compositions of continuous functions are continuous.

Example 1

Where is $h(x) = \dfrac{\sqrt{x+2}}{x-1}$ continuous?

Solution. The root needs $x \geq -2$; the denominator is 0 at $x = 1$. h is continuous on $[-2, 1) \cup (1, \infty)$.

Example 2 • From a graph



Solution. f is defined on $[-2, 4]$ except $x = 1$. It is continuous on $[-2, 1)$ and on $(1, 4]$, so on any closed interval that avoids 1 (such as $[2, 4]$), but not on $[0, 2]$.

Common mistakes

- Forgetting domain restrictions from roots and logarithms.
- Including an endpoint where the function is undefined.

Quick check

Where is each function continuous? 1. $\ln(4-x)$ 2. $\dfrac{x}{x^2+1}$ 3. $\sqrt{x}\,e^x$

Answers. 1. $(-\infty, 4)$. 2. All real numbers. 3. $[0, \infty)$.